RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA

B.Tech. First Year (I & II Semester) • Choice Based Grading System

EXAM PREPARATION GUIDE & PYQ ANALYSIS COHORT

MATHEMATICS - II (BT-202)

Compiled From:	Official PYQs (2020 – 2025)
Syllabus Structure:	Unit-Wise & Topic-Wise Analysis
Format:	Theory, Derivations, and Numericals separated
Target Intake:	June 2026 Upcoming Examinations

EXECUTIVE SUMMARY & ANALYSIS METHODOLOGY

This comprehensive preparation guide has been meticulously engineered using data from 8 cycles of Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV) examination question papers for the course **Mathematics - II (BT-202)** spanning from 2020 to 2025. Every question has been parsed, mapped to its core academic domain, evaluated for occurrence frequency, and grouped unit-wise and topic-wise to provide an optimal roadmap for student success.

Weightage and Priority Classification Strategy

To optimize study time, topics within each unit are labeled using four explicit urgency levels based on real statistical data:

- **MOST IMPORTANT**: Concepts appearing across more than 75% of past papers or carrying mandatory heavy-mark weights (e.g., Variation of Parameters, Cauchy's Integral Formula, Charpit's Method).
- **FREQUENTLY ASKED**: Highly reliable questions with regular alternate-year repetition cadence.
- **MODERATE IMPORTANCE**: Fundamental concepts that appear regularly but form single-part questions.
- **RARE BUT POSSIBLE**: Niche curriculum requirements or challenging variants that appear isolatedly.

> 💡 **CRITICAL EXAM INSIGHT**
>
> RGPV Mathematics-II exam papers follow a strict layout: 8 extensive questions are given, each usually split into two parts (a and b), worth 7 marks each, or as a single large 14-mark block. Students must attempt any 5 complete questions. Perfect mastery over 3 complete units plus core elements of a 4th guarantees maximum scoring potential.

UNIT 1: ORDINARY DIFFERENTIAL EQUATIONS (FIRST ORDER)

1.1 Topic-Wise Weightage & Priority

Topic	Frequency	Historical Marks	Priority Classification
Leibnitz's Linear & Reducible (Bernoulli's Form)	5 Times	35 Marks	MOST IMPORTANT
Exact Differential Equations & Test for Exactness	3 Times	21 Marks	FREQUENTLY ASKED
First-Order Variable Separable / Basic Substitutions	2 Times	14 Marks	MODERATE IMPORTANCE

1.2 Important Formulas & Core Patterns

1. Leibnitz's Linear Form: $dy/dx + Py = Q$.
Integrating Factor: $I.F. = e^{\int P\,dx}$. Solution: $y \cdot (I.F.) = \int (Q \cdot I.F.)\,dx + C$.

2. Bernoulli's Equation Form: $dy/dx + Py = Q y^n$.
Divide by y^n and substitute $v = y^{1-n}$ to reduce to Leibnitz form.

3. Exactness Criterion: For $M\,dx + N\,dy = 0$, it is exact if $\partial M/\partial y = \partial N/\partial x$.

1.3 Extracted Past Exam Questions

Solve $(1+y^2)dx = (\tan^{-1} y - x)dy$

Type: Numerical / Leibnitz Linear | **Occurrences:** 4 Times | **Appeared In:** Dec 2024 (1a), Dec 2023 (1a), June 2023 (1b), June 2022 (1b - layout variant)

Solve $(e^y+1)\cos x\,dx + e^y \sin x\,dy = 0$

Type: Exactness Verification / Separable | **Occurrences:** 3 Times | **Appeared In:** June 2025 (1a), Dec 2023 (1b), June 2020 (4b - explicitly asked to test for exactness)

Solve using Bernoulli's Method: $x\,dy/dx + y = x^3 y^6$

Type: Numerical | **Occurrences:** 1 Time | **Appeared In:** June 2024 (1a)

Solve using Exact method: $(x e^{xy} + 2y)dy/dx + y e^{xy} = 0$

Type: Numerical / Exact | **Occurrences:** 1 Time | **Appeared In:** June 2024 (1b)

Solve: $dy/dx = \cos(x+y) + \sin(x+y)$

Type: Numerical / Substitution | **Occurrences:** 1 Time | **Appeared In:** June 2023 (1a)

Solve using Bernoulli's Method: $dy/dx + y \tan x = y^2 \sec x$

Type: Numerical | **Occurrences:** 1 Time | **Appeared In:** June 2022 (2a)

Solve using Bernoulli's Method: $\cos x\,dy = y(\sin x - y)dx$

Type: Numerical | **Occurrences:** 1 Time | **Appeared In:** Nov 2022 (1a)

Solve the Linear differential equation: $\sin 2x\,dy/dx - y = \tan x$

Type: Numerical | **Occurrences:** 1 Time | **Appeared In:** Nov 2022 (1b)

Solve $(r + \sin\theta - \cos\theta)dr + r(\sin\theta + \cos\theta)d\theta = 0$

Type: Numerical / Exact polar | **Occurrences:** 1 Time | **Appeared In:** Nov 2022 (2a)

Solve the linear differential equation: $dy/dx + 2y/x = \sin x$

Type: Numerical | **Occurrences:** 1 Time | **Appeared In:** June 2020 (8b)

1.4 Most Expected Questions (Upcoming Exams)

1. **High Probability Numerical:** Solve $(1+y^2)dx = (\tan^{-1} y - x)dy$. It represents the single highest repeating pattern in Unit 1.
2. **Bernoulli Variant:** Solve $x\, dy/dx + y = x^3 y^6$ tracking back to recent trends.

UNIT 2: ORDINARY DIFFERENTIAL EQUATIONS (HIGHER ORDER & SPECIAL FUNCTIONS)

2.1 Topic-Wise Weightage & Priority

Topic	Frequency	Historical Marks	Priority Classification
Method of Variation of Parameters	6 Times	70 Marks	MOST IMPORTANT
Bessel Functions & Legendre Polynomials (Properties / Series)	6 Times	42 Marks	MOST IMPORTANT
Linear ODEs with Constant Coefficients (C.F. & P.I.)	7 Times	49 Marks	FREQUENTLY ASKED
Cauchy's Homogeneous / Legendre's Linear Equations	2 Times	14 Marks	MODERATE IMPORTANCE
Simultaneous Linear Differential Equations	2 Times	14 Marks	MODERATE IMPORTANCE

2.2 Important Formulas & Core Patterns

1. Variation of Parameters for Second Order $y'' + P y' + Q y = X$:
Let Complementary Function be $y_c = C_1 y_1 + C_2 y_2$. Wronskian is defined as:

$$W = y_1 y_2' - y_2 y_1'$$

The Particular Integral is $y_p = u y_1 + v y_2$, where:

$$u = - \int (y_2 X / W)\, dx, \qquad v = \int (y_1 X / W)\, dx$$

2. Bessel's Fundamental Recurrence: $d/dx\, [x^n J_n(x)] = x^n J_{(n-1)}(x)$.
3. Half-Order Bessel Identity: $J_{(1/2)}(x) = \sqrt{(2 / \pi x)} \sin x$.

2.3 Extracted Past Exam Questions

A. Method of Variation of Parameters (Always 14 Marks or Full Parts)

Solve $(D^2 + a^2)y = \tan ax$ using the method of variation of parameters.

Occurrences: 3 Times | **Appeared In:** June 2025 (3), June 2022 (3), June 2024 (Layout variant as D^2+9)

Solve $(D^2 + 1)y = x$ by variation of parameters.

Occurrences: 1 Time | **Appeared In:** Dec 2024 (2b)

Solve $(D^2 + 1)y = x \sin x$ using variation of parameters.

Occurrences: 1 Time | **Appeared In:** June 2024 (3 - 14 Marks)

Solve $(D^2 + 9)y = \tan 3x$ by variation of parameters.

Occurrences: 1 Time | **Appeared In:** Dec 2023 (3 - 14 Marks)

Solve $(D^2 + 4)y = \tan 2x$ by variation of parameters.

Occurrences: 1 Time | **Appeared In:** Nov 2022 (3)

B. Higher-Order Constant Coefficients & Homogeneous Equations

Solve $(D^2 - 5D + 6)y = 4e^x + 5$

Appeared In: June 2025 (1b)

Solve Homogeneous Equation: $x^2 \frac{d^2y}{dx^2} + 5x \frac{dy}{dx} + 4y = x \log x$

Appeared In: June 2025 (2a)

Solve $(D^2 + 3D + 2)y = \sin 3x$

Appeared In: Dec 2024 (1b)

Solve $(1+x)^2 \frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = \cos \log(1+x)$

Appeared In: Dec 2024 (3a)

Solve $(D^2 - 6D + 13)y = 8e^{3x} \sin 2x$

Appeared In: June 2024 (2a)

Solve $(D^2 - 4D + 3)y = \cos 2x$

Appeared In: Dec 2023 (2a)

Solve $d^2y/dx^2 + dy/dx = (1+e^x)^{-1}$

Appeared In: June 2023 (2a)

Solve $(D^2 - 2D - 3)y = x^3 e^{-3x}$

Appeared In: June 2022 (2b)

Solve $(D^3 - 7D^2 + 14D - 8)y = e^x \cos 2x$

Appeared In: Nov 2022 (2b)

Solve basic order equations: $d^2y/dx^2 - 4\, dy/dx + 3y = 0$ AND $(D+2)(D-1)^3 y = e^x$

Appeared In: June 2020 (1a, 1b)

Solve by changing into normal form: $d^2y/dx^2 - 2 \tan x\, dy/dx + 5y = \sec x \cdot e^x$

Appeared In: June 2020 (5a)

C. Simultaneous Differential Equations

Solve simultaneous linear systems: $dx/dt - 7x + y = 0$ and $dy/dt - 2x - 5y = 0$

Appeared In: Dec 2024 (2a)

Solve with initial boundaries: $dx/dt - y = e^t$, $dy/dt + x = \sin t$ given $x(0)=1, y(0)=0$

Appeared In: June 2023 (2b)

D. Special Functions (Bessel, Legendre, Frobenius Derivations)

Show that $J_n(-x) = (-1)^n J_n(x)$ when n is an integer.

Occurrences: 3 Times | **Appeared In:** Dec 2024 (3b), June 2020 (2b), June 2024 (Recurrence alternate)

Prove that $J_{1/2}(x) = \sqrt{2 / \pi x} \sin x$

Occurrences: 2 Times | **Appeared In:** Dec 2023 (2b), June 2023 (4a)

Solve in series Legendre's Differential Equation: $(1-x^2)y'' - 2xy' + n(n+1)y = 0$

Occurrences: 2 Times | **Appeared In:** June 2025 (2b), June 2020 (2a)

Show that $d/dx\, [x^n J_n(x)] = x^n J_{n-1}(x)$

Occurrences: 1 Time | **Appeared In:** June 2024 (2b)

Solve the differential equation $x(1-x)y'' + 2(1-2x)y' - 2y = 0$ using the Frobenius series method.

Occurrences: 1 Time | **Appeared In:** June 2023 (3 - 14 Marks)

2.4 Most Expected Questions (Upcoming Exams)

1. **Guaranteed Concept:** Solve $(D^2 + a^2)y = \tan ax$ or its numerical equivalent $(D^2 + 9)y = \tan 3x$ using Variation of Parameters.
2. **Derivation Task:** Prove that $J_{1/2}(x) = \sqrt{2/\pi x}\,\sin x$ or show the parity rule $J_n(-x) = (-1)^n J_n(x)$.

UNIT 3: PARTIAL DIFFERENTIAL EQUATIONS (PDE)

3.1 Topic-Wise Weightage & Priority

Topic	Frequency	Historical Marks	Priority Classification
Charpit's General Method for Non-Linear PDEs	5 Times	35 Marks	MOST IMPORTANT
Homogeneous / Non-Homogeneous Linear PDEs with Constant Coefficients	5 Times	35 Marks	MOST IMPORTANT
Formation of PDE by Eliminating Arbitrary Functions/ Constants	4 Times	28 Marks	FREQUENTLY ASKED
Lagrange's Linear PDE $Pp + Qq = R$	3 Times	21 Marks	FREQUENTLY ASKED

3.2 Important Formulas & Core Patterns

1. Lagrange's Auxiliary System: For $Pp + Qq = R$, form fractions $dx/P = dy/Q = dz/R$ and resolve by grouping or multipliers.

2. Charpit's Auxiliary Equations: To solve $F(x,y,z,p,q)=0$, use:

$$dp / (F_x + p F_z) = dq / (F_y + q F_z) = dz / (-p F_p - q F_q) = dx / (-F_p) = dy / (-F_q)$$

Find a relation between p and q, solve for them, then integrate $dz = p\,dx + q\,dy$.

3. C.F. for Higher Order PDEs: If $F(D, D')z = 0$ features factor $(D - m D')$, then $C.F. = \varphi(y + mx)$.

3.3 Extracted Past Exam Questions

A. PDE Formation / Elimination of Arbitrary Functions

Eliminate the arbitrary function f from relation: $z = y^2 + 2f(1/x + \log y)$

Appeared In: June 2025 (4a)

Construct partial differential equation from: $f(x^2+y^2+z^2, z^2-2xy) = 0$

Appeared In: Dec 2024 (5a)

Form partial differential equation from: $Z = (x+y)\phi(x^2-y^2)$

Appeared In: June 2024 (4a)

Form partial differential equation by eliminating function from: $z = f(y/x)$

Appeared In: June 2022 (8a)

B. Lagrange's Linear PDE Solutions ($Pp + Qq = R$)

Solve: $(y+z)p + (x+z)q = x+y$

Appeared In: June 2025 (5a)

Solve: $(x-y)p + (x+y)q = 2xz$

Appeared In: Dec 2023 (4a)

Solve: $yp - xq = z$

Appeared In: June 2024 (5a)

C. Charpit's Non-Linear PDE Method

Solve by Charpit's method: $(p^2 + q^2)y = qz$

Occurrences: 2 Times | **Appeared In:** Dec 2023 (4b), June 2023 (4b)

Solve by Charpit's method: $px + qy = pq$

Occurrences: 1 Time | **Appeared In:** Dec 2024 (4a)

Solve by Charpit's method: $q = 3p^2$

Occurrences: 1 Time | **Appeared In:** June 2020 (3a)

Solve: $x^2 p^2 + y^2 q^2 = z^2$ (Reducible to Charpit / Clariaut standard forms)

Occurrences: 1 Time | **Appeared In:** Nov 2022 (8a)

D. Higher-Order Homogeneous / Non-Homogeneous Linear PDEs

Solve: $(D^2 - DD' - 6D'^2)Z = xy$

Occurrences: 2 Times | **Appeared In:** June 2024 (4b), June 2020 (3b - listed as z instead of Z)

Solve: $d^2z/dx^2 - d^2z/dy^2 = x^2 y$

Appeared In: June 2025 (4b)

Solve: $(D^3 - 4D^2 D' + 4D D'^2)Z = \cos(2x+y)$

Appeared In: Dec 2024 (4b)

Solve: $(D^2 + 4DD' - 5D'^2)Z = \sin(2x+3y)$

Appeared In: Dec 2023 (5a)

Solve: $(D^2 - 6DD' + 9D'^2)z = 12x^2 + 36xy$

Appeared In: June 2023 (5a)

Solve: $(D^3 - 3D^2 D' + 4D'^3)z = e^{(x+2y)}$

Appeared In: June 2022 (8b)

Solve: $(D^2 - 4DD' + 4D'^2)Z = \cos(x-2y)$

Appeared In: Nov 2022 (8b)

3.4 Most Expected Questions (Upcoming Exams)

1. **Charpit Method Standard Target:** Solve $(p^2 + q^2)y = qz$. This exact mathematical formulation has repeated multiple times.
2. **Higher Order PDE P.I. calculation:** Solve $(D^2 - DD' - 6D'^2)Z = xy$ where the Particular Integral requires algebraic series expansion.

UNIT 4: COMPLEX VARIABLES (ANALYTIC FUNCTIONS & COMPLEX INTEGRATION)

4.1 Topic-Wise Weightage & Priority

Topic	Frequency	Historical Marks	Priority Classification
Analytic Functions & Cauchy-Riemann (C-R) Conditions	9 Times	63 Marks	MOST IMPORTANT
Residue Calculation at Poles & Residue Theorem	6 Times	42 Marks	MOST IMPORTANT
Cauchy's Integral Formula & Theorems	5 Times	35 Marks	FREQUENTLY ASKED

Topic	Frequency	Historical Marks	Priority Classification
Line Integrals along Paths / Contours	3 Times	21 Marks	MODERATE IMPORTANCE

4.2 Important Formulas & Core Patterns

1. Cauchy-Riemann equations (Cartesian form): For $f(z) = u + iv$ to be analytic, it requires:

$$\partial u/\partial x = \partial v/\partial y \quad \text{and} \quad \partial u/\partial y = -\partial v/\partial x$$

2. Cauchy's Integral Formula: For a pole of order n at $z = a$ inside boundary contour C:

$$oint\_C\ f(z)/(z-a)\text{^}n\ dz = [2\pi i / (n-1)!] * f\text{^}((n-1))(a)$$

3. Residues at Simple vs Multiple Order Poles:
Simple pole ($n=1$): $Res(a) = lim\_(z \to a)\ (z-a)f(z)$.
Order n pole: $Res(a) = [1 / (n-1)!] * lim\_(z \to a)\ d\text{^}(n-1)/dz\text{^}(n-1)\ [(z-a)\text{^}n\ f(z)]$.

4.3 Extracted Past Exam Questions

A. Analytic Functions, Harmonic Conjugates, and C-R Equations

Determine parameter p so that $f(z) = 1/2\ log(x\text{^}2+y\text{^}2) + i\ tan\text{^}-1(px/y)$ is analytic.
Occurrences: 3 Times | **Appeared In:** Dec 2024 (6a), Dec 2023 (5b), June 2024 (Analogue context)

Show that $u = e\text{^}-x\ (x \sin y - y \cos y)$ is Harmonic.
Occurrences: 2 Times | **Appeared In:** Dec 2024 (5b), June 2024 (5b)

Construct the analytic function $f(z)$ whose real part is $e\text{^}x \cos y$.
Occurrences: 2 Times | **Appeared In:** June 2022 (6b), Dec 2023 (6a variant - verify harmonic and get conjugate)

Find all values of K such that $f(z) = e\text{^}x(\cos ky + i \sin ky)$ is analytic.
Occurrences: 1 Time | **Appeared In:** June 2025 (5b)

Prove that an analytic function with a constant modulus must be a constant function.
Occurrences: 1 Time | **Appeared In:** June 2023 (5b)

Prove the operator derivative identity: $(\partial\text{^}2/\partial x\text{^}2 + \partial\text{^}2/\partial y\text{^}2)|f(z)|\text{^}2 = 4|f'(z)|\text{^}2$
Occurrences: 1 Time | **Appeared In:** June 2022 (6a)

Show that $f(Z) = z\quad ar\{z\}$ is differentiable but not analytic at the origin.
Occurrences: 1 Time | **Appeared In:** Nov 2022 (6a)

Show that $u(x,y) = e^{-2x} \sin 2y$ is harmonic and find its conjugate.

Occurrences: 1 Time | **Appeared In:** Nov 2022 (6b)

Show that the function $f(z) = e^z$ is analytic everywhere.

Occurrences: 1 Time | **Appeared In:** June 2020 (6a)

Show that the function $u = e^{(-2ny)} \sin(x^2-y^2)$ is harmonic.

Occurrences: 1 Time | **Appeared In:** June 2020 (7b)

Determine whether functions $1/z$ or basic options are analytic.

Occurrences: 1 Time | **Appeared In:** June 2020 (7a)

B. Complex Integration & Cauchy Formula Evaluations

Evaluate line integral $\int (3x^2 + 4xy + i\,x^2)\,dz$ along parabolic arc $y = x^2$ from (0,0) to (1,1).

Occurrences: 2 Times | **Appeared In:** June 2025 (6a), June 2024 (6a)

Evaluate using Cauchy's Theorem/Formula: $\int_c z^3 e^{-z} / (z-1)^3\,dz$ inside boundary circle $|z-1| = 1/2$.

Occurrences: 1 Time | **Appeared In:** Dec 2024 (6b)

Use Cauchy Integral formula to solve: $\oint_C (\sin \pi z^2 + \cos \pi z^2) / ((z-1)(z-2))\,dz$ where contour C is circle $|z| = 3$.

Occurrences: 1 Time | **Appeared In:** June 2023 (6a)

Solve using complex integration methods: $\int_0^{2\pi} \cos 4\theta / (5 + 4 \cos \theta)\,d\theta$

Occurrences: 1 Time | **Appeared In:** June 2023 (6b)

Solve piece-wise: $\int (x - y + i\,x^2)\,dz$ from $z=0$ to $z=1$ along real axis, then to $1+i$ along imaginary shift.

Occurrences: 1 Time | **Appeared In:** June 2023 (7a)

Using Cauchy's formula, calculate $\int_c e^{(2z)} / (z+1)^3\,dz$ where C: $|z|=2$.

Occurrences: 1 Time | **Appeared In:** June 2022 (7a)

Evaluate contour integral $\int_c 1 / ((z+4)z^8)\,dz$ along circle boundary $|z|=2$.

Occurrences: 1 Time | **Appeared In:** June 2022 (7b)

Using Cauchy integral theorem, evaluate $\int_c e^{(2z)} / ((z-1)^2(z-3))\,dz$ with contour circle $|Z|=2$.

Occurrences: 1 Time | **Appeared In:** Nov 2022 (7b)

Evaluate contour integral $\oint_c z / (z^2 + 9)\, dz$ where contour is circle $|z-2i|=4$.

Occurrences: 1 Time | **Appeared In:** June 2020 (6b)

C. Poles & Residue Computations

If $f(z) = 1 / ((z-1)(z-2)^4)$, compute residues at all poles.

Occurrences: 1 Time | **Appeared In:** June 2025 (6b)

Find the poles and residues at each pole of $e^{iz} / (z^2+1)$.

Occurrences: 1 Time | **Appeared In:** Dec 2024 (7a)

Find the Poles and Residues at each pole of $f(z) = \sin^2 z / (z - \pi/6)^2$.

Occurrences: 1 Time | **Appeared In:** June 2024 (6b)

Find the residue of $z e^z / (z-1)^3$ at its pole.

Occurrences: 2 Times | **Appeared In:** Dec 2023 (6b), Standard text core matches

By Residue theorem, evaluate $\int_c \tan z / (z^2 - 1)\, dz$ where contour $C: |z|=2$.

Occurrences: 1 Time | **Appeared In:** Nov 2022 (7a)

4.4 Most Expected Questions (Upcoming Exams)

1. **High Reliability Analytic Check:** Determine p such that $f(z) = 1/2 \log(x^2+y^2) + i \tan^{-1}(px/y)$ is analytic. This question has run through continuous cycles.
2. **Residue Calculus:** Find residues for functions with multiple order poles like $z e^z / (z-1)^3$.

UNIT 5: VECTOR CALCULUS (VECTOR THEOREMS & MULTI-INTEGRALS)

5.1 Topic-Wise Weightage & Priority

Topic	Frequency	Historical Marks	Priority Classification
Green's Theorem in Plane Boundaries	4 Times	42 Marks	MOST IMPORTANT
Gauss Divergence Theorem (Cube Contours)	3 Times	42 Marks	MOST IMPORTANT
Directional Derivatives & Scalar Potentials	4 Times	28 Marks	FREQUENTLY ASKED
Stokes Theorem (Box / Vector Verification)	3 Times	28 Marks	FREQUENTLY ASKED

Topic	Frequency	Historical Marks	Priority Classification
Vector Identities ($r^n \vec{r}$ properties)	4 Times	28 Marks	FREQUENTLY ASKED

5.2 Important Formulas & Core Patterns

1. Vector Operator Identities for Position Vector $\vec{r} = x\,i + y\,j + z\,k$:

$$\text{div}(\vec{r}) = 3 \quad \text{and} \quad \text{curl}(\vec{r}) = \vec{0}$$

$$\text{grad}(f(r)) = f'(r) * (\vec{r} / r)$$

$$\text{div}(r^n \vec{r}) = (n+3)r^n \Rightarrow \text{Solenoidal if } n = -3$$

$$\text{curl}(r^n \vec{r}) = \vec{0} \Rightarrow \text{Irrotational for any real } n$$

2. Green's Theorem: $\oint_C (M\,dx + N\,dy) = \int_R (\partial N/\partial x - \partial M/\partial y)\,dx\,dy$.

3. Gauss Divergence Theorem: $\oint_S \vec{F} \cdot d\vec{S} = \int_V (\text{div } \vec{F})\,dV$.

4. Stokes Theorem: $\oint_C \vec{F} \cdot d\vec{r} = \int_S (\text{curl } \vec{F}) \cdot d\vec{S}$.

5.3 Extracted Past Exam Questions

A. Vector Identities & Differential Operations

If $\vec{r}$ is the position vector of any point in space, prove that $r^n \vec{r}$ is irrotational.

Occurrences: 2 Times | **Appeared In:** June 2025 (7a), Dec 2023 (8a - written as $\text{curl}(r^n \vec{r}) = \vec{0}$)

Prove that $r^n \vec{r}$ is Solenoidal if $n = -3$.

Occurrences: 2 Times | **Appeared In:** June 2022 (4b), Nov 2022 (4a variant as $\vec{r}/r^3$)

Prove the laplacian identifier identity: $\nabla^2 f(r) = f''(r) + (2/r)f'(r)$

Occurrences: 1 Time | **Appeared In:** June 2023 (7b)

Show that vector $(x^2 - yz)i + (y^2 - zx)j + (z^2 - xy)k$ is Irrotational and calculate its scalar potential.

Occurrences: 1 Time | **Appeared In:** Nov 2022 (4b)

Find $\text{div}(\text{curl } F)$ where $F = x^2 y\,i + y\,z\,j + 2yz\,k$.

Occurrences: 1 Time | **Appeared In:** June 2020 (8a)

B. Directional Derivatives & Line Work Integrals

Find the workdone by the force field $F = z\,i + x\,j + y\,k$ as it moves a particle along the arc curve $r\_vec = \cos t\, i + \sin t\, j - t\, k$ from $t=0$ to $t=2\pi$.

Appeared In: June 2025 (7b)

Find directional derivative of $\varphi = x^2 y z + 4 x z^2$ at coordinates $(1, -2, -1)$ along direction vector $2i - j - 2k$.

Appeared In: Dec 2024 (7b)

Find directional derivative of $f(x,y,z) = x y^2 + y z^3$ at $(2, 1, 1)$ in vector direction $i + 2j + 2k$.

Appeared In: June 2024 (8a)

Find directional derivative of $f(x,y,z) = e^{(2x)} \cos yz$ at origin $(0,0,0)$ along tangent direction to curve $x = a \sin t, y = a \cos t, z = at$ evaluated at $t = \pi/4$.

Appeared In: June 2023 (8a)

Find the angle between intersect surfaces $x^2+y^2+z^2=9$ and $z = x^2+y^2-3$ at coordinates $(2, -1, 2)$.

Appeared In: June 2022 (4a)

C. Integral Theorems (Green, Gauss, Stokes Verification - 14 Marks)

Verify Green's Theorem for $\int_C [(xy+y^2)dx + x^2 dy]$ where boundary C is defined by curves $y = x$ and $y = x^2$.

Occurrences: 2 Times | **Appeared In:** Dec 2024 (8), Standard cross-checks

Verify Stokes Theorem for $F = (x^2 - y^2)i + 2xy\, j$ over flat rectangular boundaries bounded by planes $x=0, x=a, y=0, y=b$.

Occurrences: 1 Time | **Appeared In:** June 2025 (8)

Verify Green's Theorem in plane for $\int_C (x^2 - x y^3)dx + (y^2 - 2xy)dy$ where contour C represents square bounds with vertices $(0,0), (2,0), (2,2), (0,2)$.

Occurrences: 1 Time | **Appeared In:** June 2024 (7)

Verify Gauss Divergence Theorem for vector field $F = x^3\, i + y^3\, j + z^3\, k$ evaluated over entire cubic space volume bounded by $x=0,a; y=0,a; z=0,a$.

Occurrences: 1 Time | **Appeared In:** Dec 2023 (7)

Using Green's Theorem, find area of region in first quadrant bounded by curves $y=x, y=1/x, y=x/4$.

Occurrences: 1 Time | **Appeared In:** June 2023 (8b)

Verify Gauss Divergence Theorem for vector field $F = x^2 i + y^2 j + z^2 k$ over cube space cross-section bounded by $x=0,a; y=0,b; z=0,c$.

Occurrences: 1 Time | **Appeared In:** June 2022 (5)

Verify Green's Theorem for $\int_c [(3x^2 - 8y^2)dx + (4y - 6xy)dy]$ where area contour C is enclosed by $x=0, y=0, x+y=1$.

Occurrences: 1 Time | **Appeared In:** Nov 2022 (5)

5.4 Most Expected Questions (Upcoming Exams)

1. **Guaranteed 14-Mark Verification:** Verification of Green's theorem using bounds $y=x$ and $y=x^2$, or Gauss Divergence over a standard cube.
2. **Differential Property Proof:** Prove that $curl(r^n \vec{r}) = \vec{0}$. It remains a core conceptual question.

STRATEGIC BLUEPRINT FOR HIGH SCORING

Overall Crucial High-Scoring Topics (Ranked by Return-on-Investment)

1. **Variation of Parameters (Unit 2):** Always appears as a major part or full question. Mastering this single method guarantees 14 marks.
2. **Vector Integrals Verification (Unit 5):** Green's or Gauss divergence theorem verification questions are long but straightforward to solve.
3. **Cauchy-Riemann & Parameter Analytics (Unit 4):** Finding analytic parameter p or computing simple contour integrals via residue rules is highly systematic.
4. **Leibnitz Linear ODE Form (Unit 1):** The specific equation $(1+y^2)dx = (\tan^{-1} y - x)dy$ has historically high repetition.

Last-Minute Revision Essentials (The 2-Hour Strategy)

- Review the formula for Wronskian and particular integrals for Variation of Parameters.
- Memorize the Charpit's auxiliary setup expression.
- Review the identity for half-order Bessel functions ($J_{1/2}$) and standard $r^n \vec{r}$ properties.
- Review the Cartesian form of C-R equations.

Predicted Paper Blueprint Questions for Upcoming Examination

- **Q1 (Linear ODE):** Solve $(1+y^2)dx = (\tan^{-1} y - x)dy$.
- **Q2 (Parameters Variation):** Solve $(D^2 + a^2)y = \tan ax$.
- **Q3 (Charpit PDE):** Solve $(p^2 + q^2)y = qz$.
- **Q4 (Complex Analytics):** Find analytic variable matching parameter p in $\frac{1}{2} \log(x^2+y^2) + i \tan^{-1}(px/y)$.
- **Q5 (Integral Proof):** Verify Green's theorem in a plane enclosed by boundaries $y=x$ and $y=x^2$.